

BACHELOR OF INFORMATION TECHNOLOGY ENGG. EXAMINATION, 20264th year, 1st semester**Data Warehousing and Data Mining**

Time: 3 Hours

Full Marks: 100

Read the questions carefully and answer accordingly.

CO1: Answer any 3 questionsFull marks: **15**

1. a) What are the major issues in data mining? Define with suitable examples: nominal attributes and ordinal attributes. (2+3)
- b) What is ETL? Explain different data extraction strategies of ETL. (1+4)
- c) Why do we need a separate Data Warehouse? Describe the multi-tier architecture of Data Warehouse with proper schematic diagram. (1+4)
- d) Describe the working procedure(s) of the following OLAP operations with proper example: drill-up, roll-down, and slice-dice. (5)

CO2: Answer any 2 questionsFull marks: **20**

- 2.a) Discuss about the Issues to be considered during Data Integration. What is the curse of dimensionality? How can it be reduced? (4+2+4)
- b) i. What is Entropy-based binning method?
ii. Consider the following data for the attribute X, use smoothing by bin means method to smooth these data, using a bin depth of 3. (4 + 6)
X = {7,12,5,8,5,9,13,12,19,7,12,12,13,3,4,5,13,8,7,6}
- c) Explain Noisy Data. What is an outlier? Find out the outliers w.r.t. the data set, D = {2.37, 2.16, 14.82, 1.73, 41.04, 0.23, 1.32, 2.91, 39.41, 0.11, 27.44, 4.51, 60.05, 0.51, 4.50, 0.18, 14.68, 4.66, 1.30, 2.06, 49.09, 1.19} using Boxplots or Histogram Analysis. (2+2+6)

CO3: Answer all questionsFull marks: **20**

3. a) Write a short note on: Market basket analysis.
- b) How to compute confidence for an association rule $X \rightarrow Y$?
- c) Find the frequent itemsets and strong association rules for the following transactional database table using Apriori algorithm. Consider the thresholds as support = 30% and confidence = 40%. Use the given dataset.
- d) Compare Apriori and FP-Growth algorithms. (3+3+10+4)

TID	items_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

CO4: Answer any 1 questionFull marks: **20**

4. a) Briefly explain classification problems and general approaches to solve them. The following table consists of training data from an employee database.

Assuming “buys_computer” as class label attribute, select appropriate splitting attribute and draw the decision tree. How can you measure that your selected splitting attribute is the best? **(4+10+6)**

RID	age	income	student	credit_rating	buys_computer
1	senior	high	yes	fair	yes
2	youth	medium	no	excellent	yes
3	middle	high	no	excellent	no
4	senior	low	no	fair	yes
5	youth	low	yes	fair	no
6	senior	medium	yes	excellent	yes
7	middle	low	no	excellent	no
8	youth	high	yes	fair	no

- b) i. What are centroid and medoid? Discuss any distance based classification algorithm with example.
 ii. What is discretization? When is it used?
 iii. Discuss the relationship between entropy and information gain.

iv. What is a confusion matrix? Explain with one example.

(2+8+5+5)

CO5: Answer any 2 questions

Full marks: 20

5. a) What are the advantages of PAM algorithm over k-means algorithm?

The given table consists of twelve objects (i.e., points) in 2-D system. Find two best possible clusters formed by the objects using the k-medoids algorithms. **(3+7)**

- b) List out all hierarchical methods for clustering data. Use Agglomerative Clustering method for finding best possible clusters of the Ice Cream Sales data set. **(3+7)**
 c) What is the drawback of k-means algorithm? Explain any density-based clustering algorithm in detail. **(2+8)**

Record	Temperature	IceCreamSales
1	18	120
2	20	150
3	22	165
4	34	320
5	40	400
6	26	210
7	28	240
8	32	300
9	24	180
10	36	350
11	30	260
12	38	370

CO6: Answer all questions

Full marks: 5

6. Write Short notes on (any one): a) “Bag of words” in text mining
 b) Mining Time-Series Data

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- CO1: Relate the features of various fundamental concepts of data warehousing like, Data Mart, Data Cube, OLAP and its variants in relation to real world problems. (K3)
 CO2: Explain and illustrate the steps of data preprocessing. (K2)
 CO3: Describe the basic algorithms of mining frequent itemsets, association rules and apply them in multilevel and multidimensional space with and without constraints (K3)
 CO4: Illustrate various approaches of classification and prediction, and relate them to specific data mining problems (K4).
 CO5: Explain various approaches of clustering and outlier detection techniques and relate them to related data mining problems (K3).
 CO6: Discuss approaches of mining in different application areas such as text, biological data, time series data. (K2)